Cross-disciplinary applications of the Toulbar2 solver

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Abstractz

The valued constraint satisfaction problem (VCSP) or weighted constraint satisfaction problem (WCSP) is a generalization of the classic CSP. We try to find an assignment of minimal cost to a set of variables given soft and hard constraints. It is usually solved using a branch-and-bound search (BnB).

By finding a good lower bound to a given instance, subtrees of the BnB can be quickly pruned to efficiently find an optimal solution. This paper studies the cross-disciplinary application of the solver Toulbar2, ranging from solving difficult combinatorial problems, to real-world application such as bioinformatics and resource management and finally artificial intelligence. In many cases where the problem was able to be written using the VCSP framework, Toulbar2 was able to outperform other state-of-the-art solvers.

Combinatorial problem solving

Cluster editing for graphs

Zahn's decision problem can be described as the cluster editing problem. Given a graph, what is the minimal number of operations (adding edges, removing edges) needed to create a disjoint union of cliques. Below an exemple of the cluster editing problem:

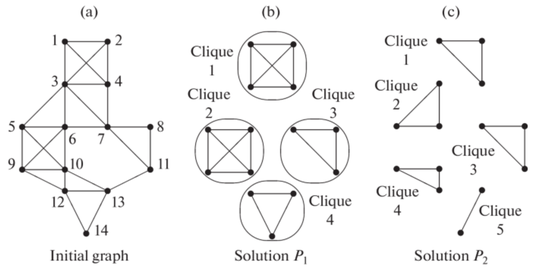


Fig. 1: Example of SMC: Robust Supply Chain with Different Network Size.

By transforming the cluster editing problem into a VCSP and solving it using Toulbar2, Mhamdi and Naanaa (2014) obtained results much better than the fixed parameter algorithm for both sparse and dense graphs.

A complicated chess problem

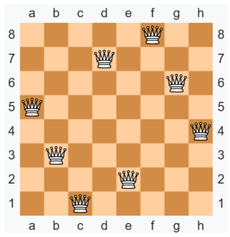


Figure 1: 8-Queens example

The N-Queen problem is a non-trivial mathematical problem that looks for a placement of N queens on a chessboard of size N by N where no two queens are threatening each other. The figure to the left shows an example of a solution where $n = 8$. The weighted N-Queen problem is the difficult optimization problem of minimizing the cost of the N-Queen if every tile incurs a cost. By modeling the weighted N-Queen problem as a VCSP, Sewa et al. (2022) demonstrated that Toulbar2 significantly outperforms other state-of-the-art solvers.

Other problems such as the quadratic assignement problem, the knapsack problem with conflict graph and discrete integrations were also solved using the VCSP framework and showed good performance of the Toulbar2 solver.

Bioinformatics & Life Sciences

Computational Protein Design (CPD)

CPD aims to create artificial proteins by identifying the amino acid sequence that minimizes the protein's energy, known as the Global Minimum Energy Conformation (GMEC). This is an NP-hard optimization problem due to the vast search space created by the numerous possible side-chain conformations, called rotamers.

Modeling with CFNs and Toulbar2

To solve this, the protein structure is mapped into a Cost Function Network (CFN): residues become variables, rotamers become values, and physical interactions are encoded as costs. Toulbar2 uses advanced branch-and-bound and pruning techniques not only to find the exact GMEC, but also to enumerate near-optimal solutions to analyze the complex fitness landscape of the protein.

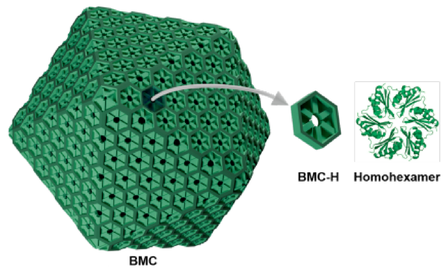


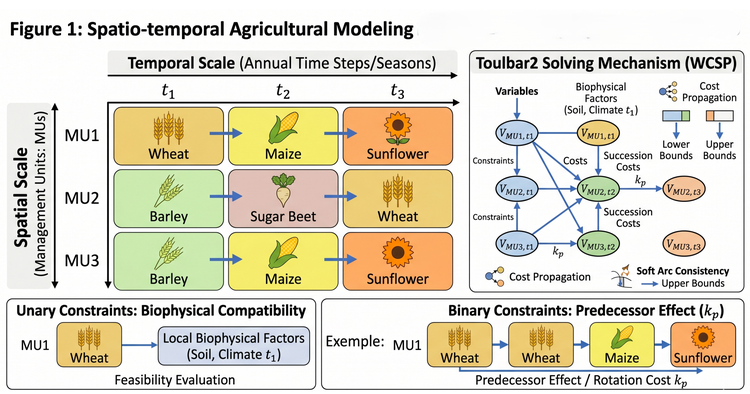
Fig. 2: BMC-H homohexamer of bacterial microcompartment (BMC) shell.

Hybrid Deep Learning Approaches

Recently, hybrid approaches have emerged to tackle multi-state design. By combining deep learning models (which produce scoring functions) with Toulbar2's exact optimization under explicit constraints, researchers can successfully design complex, symmetrical multi-component proteins—like the homohexamer shown above—while simultaneously avoiding undesirable negative states.

Recherche Opérationnelle & Planification

Le formalisme $WCSP$ performe dans la modélisation d'environnements complexes en combinant des règles strictes (contraintes dures) et l'optimisation de critères de qualité (contraintes souples).



Artificial Intelligence & Machine Learning

Toulbar2 struggles with real world applications since exhaustive search creates computational bottlenecks on large instances. We can integrate machine learning, introducing new ways to improve the solution quality. For example, algorithms like KOCO SMC handle large instances by tracking probability bounds during the inference process. This early evaluation prevents the solver from performing exhaustive enumerations.

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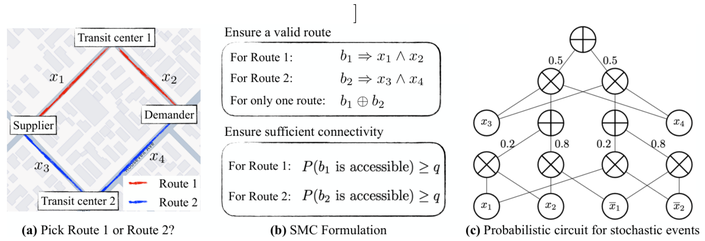

(a) Pick Route 1 or Route 2? **(b)** SMC Formulation **(c)** Probabilistic circuit for stochastic events

Fig. 3: Example of SMC: Robust Supply Chain with Different Network Size.[cite: 1072

Deep learning models can also transform raw sensory data directly into Graphical Models. This integration merges the perception of neural networks with the exact reasoning of constraint solvers to increase overall flexibility. For example, a neural network can translate unmapped protein structures or raw images into cost matrices, allowing Toulbar2 to focus exclusively on executing the exact discrete optimization.

Conclusion

The was to understand how a solver based on CFNs, such as Toulbar2, established itself across such diverse disciplines. The answer lies in combining the expressiveness of the WCSP formalism with the power of its local consistency algorithms (EDAC,VAC, OSAC).

This demonstrates that Toulbar2 has continuously evolved to adapt to different problems. It first became essential in bioinformatics for processing dense molecular networks. Then proved its competitiveness in resource planning by integrating sophisticated global constraints.

We've also seen that to what extent can constraint solvers like Toulbar2 be integrated into the training of deep learning model to correct their logical errors. In the longer term, we may witness the emergence of a standardized hybrid AI.

References

Key References:

Li, J., Jiang, N., Xue, Y. (2025). Solving Satisfiability Modulo Counting Exactly with Probabilistic Circuits. arXiv:250
1072]